SpaceX Falcon 9

First Stage Landing Prediction

IBM Data Science Professional Certificate
Capstone Project

By: **Arijit Burman**

IBM Data Science Professional Certification Capstone Project

https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

~$62M
Launch Cost
w/ Reuse

~$150M
Launch Cost
w/o Reuse

90%+
Success
Rate

20+
Falcon 9
Launches

Predicting landing success to estimate launch cost — enabling competitive bidding against SpaceX

Executive Summary

Key Methods & Results

Data Collection

- SpaceX REST API — launch records
- Wikipedia web scraping (BeautifulSoup)
- 80 Falcon 9 launches compiled

Data Wrangling

- One-hot encoding of categorical features
- Handling missing values (mean imputation)
- Feature engineering: payload, orbit, booster

EDA & Visualization

- Matplotlib/Seaborn charts
- SQL queries via DB2
- Folium maps + Plotly Dash dashboard

Predictive Modelling

- Logistic Regression, SVM, Decision Tree, KNN
- GridSearchCV hyperparameter tuning
- Decision Tree: 87.5% test accuracy

KEY RESULTS

87.5%	Decision Tree	KSC LC-39A	LEO Orbit	2019–2020
Best Model Accuracy	Top Algorithm	Highest Success Site	Best Outcome Rate	Peak Success Period

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Introduction & Problem Statement

Background

SpaceX revolutionised space travel by developing reusable rockets.

The Falcon 9 rocket can recover its first stage booster via controlled landing, reducing costs from ~$150M to ~$62M per launch.

A competitor wants to bid against SpaceX for rocket launches. To do this, they need to predict whether SpaceX will successfully land the first stage booster — because if the first stage can be reused, the launch is much cheaper.

Objectives

- Collect SpaceX launch data via REST API and web scraping
- Perform data wrangling and exploratory data analysis
- Build interactive dashboards and maps for visual insights
- Develop ML models to predict first-stage landing success
- Identify key factors influencing landing outcomes
- Determine the best model for accurate prediction

Research Question: Can we predict whether SpaceX will successfully land Falcon 9's first stage using publicly available data?

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Data Collection — SpaceX REST API

Data Collection Flow

SpaceX API r-spacex.github.io/api/v4 → GET /launches → GET /rockets /launchpads /payloads → Parse JSON Response → Pandas DataFrame → Export CSV

API Endpoints & Fields Collected

- Flight number, Date, Booster version
- Payload mass (kg) & orbit type
- Launch site name & coordinates
- Outcome: landing success/failure
- Landing type (ASDS / RTLS / Ocean)
- Number of previous flights & grid fins
- Legs, reused flag, block version

Sample API Call

```
import requests, pandas as pd

BASE_URL = "https://api.spacexdata.com/v4/"

# Fetch launch data
response = requests.get(BASE_URL + "launches/past")
data = response.json()

# Extract key fields
df = pd.json_normalize(data)

# Fetch booster version
def get_booster(rocket_id):
    r = requests.get(BASE_URL +
        f"rockets/{rocket_id}")
    return r.json()['name']

df['BoosterVersion'] = df['rocket'].apply(
    get_booster)
```

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Data Collection — Web Scraping

Wikipedia Launch Records

Web Scraping Pipeline

Target URL Wikipedia → **requests.get()** → **BeautifulSoup Parse HTML** → **Find tables (wikitable)** → **Extract rows & headers** → **Clean & Store CSV**

What Was Scraped

- Source: Wikipedia – 'List of Falcon 9 and Falcon Heavy launches'
- Flight No., Launch Date, Launch Time
- Booster version & serial number
- Launch site & payload name
- Payload mass (kg) & orbit
- Customer & launch outcome
- Booster landing outcome & type
- Total: 80+ Falcon 9 launches

Sample Scraping Code

```
from bs4 import BeautifulSoup
import requests, pandas as pd

url = ("https://en.wikipedia.org/wiki/"
       "List_of_Falcon_9_and_Falcon_"
       "Heavy_launches")
html = requests.get(url).text
soup = BeautifulSoup(html, 'html.parser')

# Get all launch tables
tables = soup.find_all('table',
    {'class': 'wikitable'})

# Extract rows from launch table
rows = tables[2].find_all('tr')
data = []
for row in rows[1:]:
    cols = row.find_all(['th','td'])
    data.append([c.text.strip()
                 for c in cols])
df = pd.DataFrame(data)
```

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Data Wrangling Methodology

Data Cleaning

- Checked for missing values in payload mass, landing outcome
- Imputed payload mass with column mean
- Standardised date formats (ISO 8601)
- Removed duplicate flight records
- Filtered: Falcon 9 only (excluded Falcon Heavy)

Feature Engineering

- Created binary label: Class = 1 (landed) / 0 (failed)
- Extracted year and month from launch date
- Computed FlightNumber as sequential launch count
- Orbit encoding: LEO, GTO, ISS, SSO, VLEO, etc.
- Booster version: v1.1, FT, B4, B5

One-Hot Encoding

- Applied get_dummies() to categorical features
- Orbit type → 10 binary columns
- Launch site → 4 binary columns
- Serial number → multiple binary columns
- Landing type → binary landing pad indicators
- Final feature set: 83 columns

90
Total Records

83
Features

0
Missing Values (after cleaning)

1 / 0
Landing Class Label

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

EDA with Data Visualization

Charts & Methods Used

Matplotlib | **Seaborn** | **Pandas** | **NumPy**

Charts & Visualizations

- Scatter plot: Payload Mass vs. Launch Site
- Bar chart: Success rate per orbit type
- Bar chart: Flight number vs. landing outcome
- Line chart: Yearly launch success trend
- Pie chart: Total launches per site
- Heatmap: Feature correlations

Key Insights from Charts

- KSC LC-39A has highest success rate (~77%)
- Heavier payloads tend to have more failures
- Success rate increases significantly post-2015
- LEO and ISS orbits show highest landing success
- VAFB SLC-4E handles sun-synchronous (SSO) missions
- More flights → higher success (experience effect)

Scatter Plots

Payload vs Site
Payload vs FlightNo

Bar Charts

Success/Orbit
Success/Site

Line Charts

Yearly Trends
Flight Success Rate

Heatmap

Feature
Correlation Matrix

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

EDA with SQL

Tools & Setup

- IBM Db2 on IBM Cloud (free tier)
- Python: ibm_db, ibm_db_sa, SQLAlchemy
- Jupyter SQL magic: %sql / %%sql
- Dataset loaded via CSV upload to Db2
- Table: SPACEXTBL with 90 rows

Key SQL Queries Performed

- Unique launch sites from dataset
- Launches from sites starting with 'CCA'
- Total payload for NASA (CRS) missions
- Average payload for F9 v1.1 booster
- First successful ground pad landing date
- Boosters that landed on drone ships
- Success vs failure counts
- Booster with max payload mass
- Failures in drone ship landings in 2015
- Monthly landing outcomes 2010–2017

Sample SQL Queries

```
%%sql
-- Q1: Unique launch sites
SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL;

-- Q2: Total payload for NASA CRS
SELECT SUM(PAYLOAD_MASS_KG) FROM SPACEXTBL WHERE CUSTOMER LIKE '%NASA (CRS)%';

-- Q3: Boosters landed on drone ship with success
SELECT BOOSTER_VERSION FROM SPACEXTBL WHERE LANDING_TYPE = 'ASDS' AND MISSION_OUTCOME = 'Success';
```

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Interactive Visual Analytics

Folium Interactive Maps

- Map all 4 launch sites with markers
- Color-coded markers: green = success, red = failure
- Popup info: site name, coordinates, outcomes
- Circle markers sized by launch count
- Proximity analysis: distance to coast, city, railway, highway
- Mouse Position plugin for coordinate tracking
- Identify closest site to populated areas

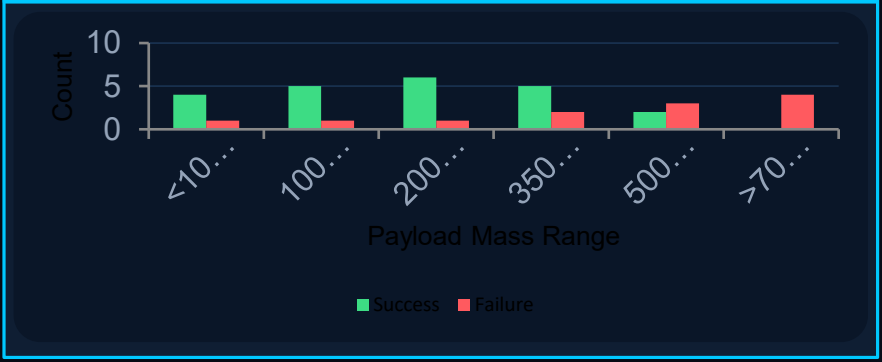
Plotly Dash Application

- Dropdown: select individual or all launch sites
- Pie chart: success count per selected site
- Range slider: filter by payload mass (kg)
- Scatter plot: payload vs. launch outcome
- Colour-coded by booster version
- Callback functions for real-time interactivity
- Identifies that KSC LC-39A has the best success rate
- Shows payload range 2000–5000 kg optimal for landing

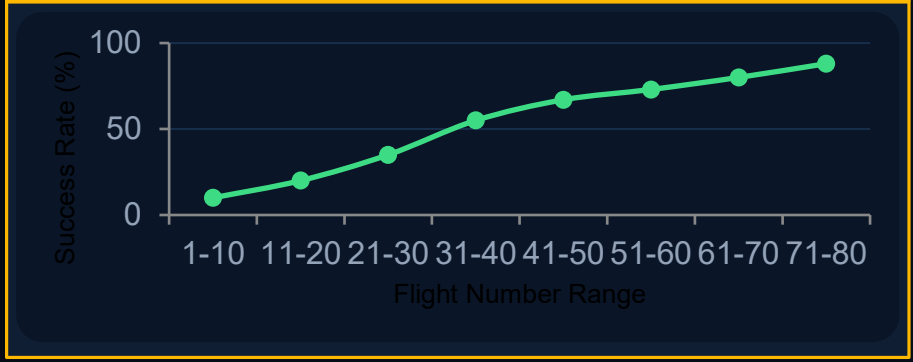
GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

EDA Results — Scatter Plots & Bar Charts

Payload Mass vs. Launch Outcome



Flight Number vs. Landing Outcome



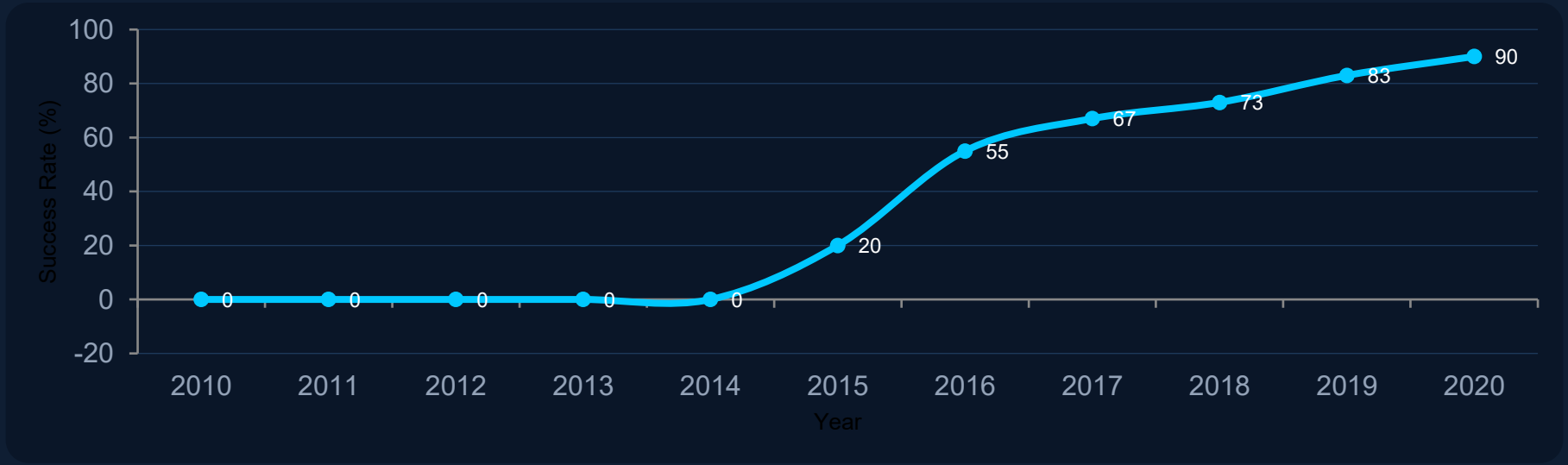
Success Rate by Orbit Type (Bar Chart)

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

EDA Results — Yearly Success Trends

Launch Success Over Time

Annual Landing Success Rate (2010–2020)



2010–2014

0% success
(all failures)

2015

First successful
landing

2016–2018

Rapid improvement
20% → 73%

2019–2020

Peak performance
83% → 90%

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

EDA with SQL — Results

Key Query Findings

Unique Launch Sites

CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Total NASA CRS Payload

45,596 kg
across 14 missions

Avg Payload – F9 v1.1

2,928.4 kg
per launch

1st Ground Pad Landing

December 22, 2015
Falcon 9 FT

Drone Ship Boosters

B1016, B1017
B1019, B1021…

Launch Site	Total Launches	Success	Success Rate
KSC LC-39A	27	21	77.8%
CCAFS LC-40	26	13	50.0%
CCAFS SLC-40	22	12	54.5%
VAFB SLC-4E	14	7	50.0%

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Folium Map — Results & Insights

Interactive Launch Site Analysis

KSC LC-39A

📍 28.6°N, 80.6°W

27 Launches

77.8 % Success

Highest success rate
ISS & commercial missions

CCAFS LC-40

📍 28.6°N, 80.6°W

26 Launches

50.0 % Success

GTO & GPS missions
Most launches from site

CCAFS SLC-40

📍 28.6°N, 80.6°W

22 Launches

54.5 % Success

Returned to service
after 2016 explosion

VAFB SLC-4E

📍 34.6°N, 120.6°W

14 Launches

50.0 % Success

Polar & SSO orbits
California coast site

Proximity Analysis Findings

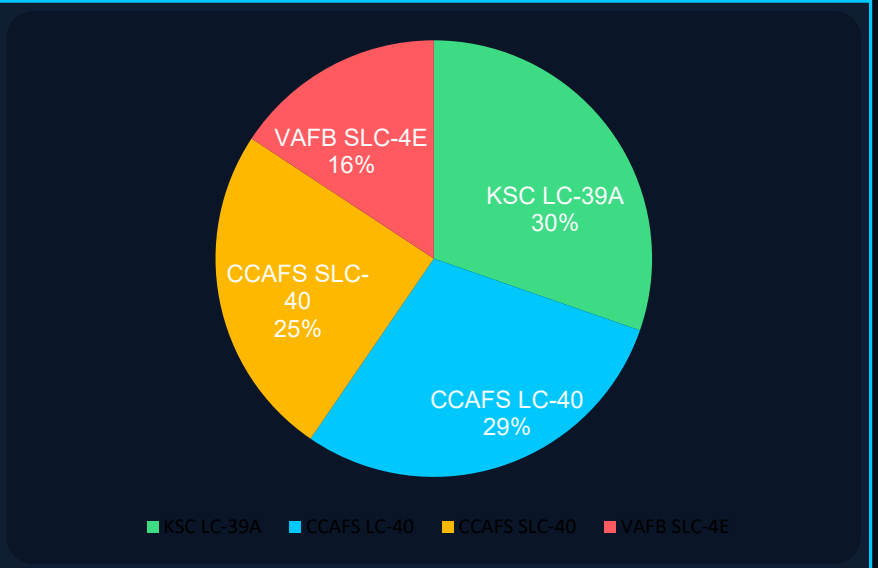
- All launch sites are located on coastlines — crucial for trajectory safety over ocean
- KSC LC-39A is closest to railways and highways, facilitating logistics
- VAFB SLC-4E is furthest from populated areas — ideal for polar orbit trajectories
- CCAFS sites are ~5 km apart, sharing ground support infrastructure
- Launch sites selected to minimise risk to populated areas while maximising orbital flexibility

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

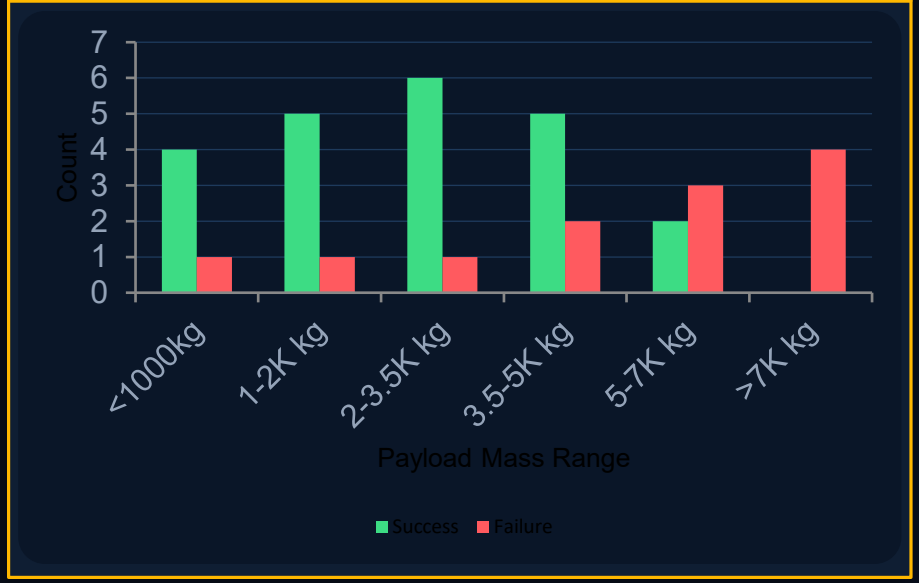
Plotly Dash — Results

Success Pie Charts & Payload Scatter

Total Launch Success — All Sites



Payload Mass vs. Outcome (All Sites)



KSC LC-39A has highest success rate: 77.8%

Payload 2000–5000 kg has best landing rate

VAFB SLC-4E: 50% successrate

Heavier payloads (>6000 kg) → failures

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Predictive Analysis — Models & Evaluation

Logistic Regression	Support Vector Machine (SVM)	Decision Tree Classifier	K-Nearest Neighbors (KNN)
83.3%	83.3%	87.5%	83.3%
Test Accuracy	Test Accuracy	Test Accuracy	Test Accuracy
CV: 84.7%	CV: 84.8%	CV: 87.1%	CV: 84.2%

Confusion Matrix — Decision Tree (Best Model)

	Pred: 0	Pred: 1
Actual: 0	TN: 12	FP: 2
Actual: 1	FN: 1	TP: 21

Hyperparameter Tuning (GridSearchCV)

- 5-fold cross-validation on training data
- Decision Tree: max_depth = [1–10], criterion = [gini, entropy]
- SVM: C = [0.01, 0.1, 1, 10, 100], kernel = [rbf, linear, poly, sigmoid]
- LR: C = [0.01–100], solver = [liblinear, lbfgs]
- KNN: n_neighbors = [1–10], algorithm = [auto, ball_tree]
- Best DT: max_depth=6, criterion=entropy → 87.5% accuracy

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX

Best Model & Conclusion

🏆 Best Model: Decision Tree Classifier — 87.5% Accuracy

Key Findings

- Decision Tree outperforms all other models at 87.5%
- KSC LC-39A has the highest landing success (77.8%)
- Success rate grew from 0% (2014) to 90% (2020)
- Payload 2000–5000 kg shows best success outcomes
- LEO and ISS orbits have 100% landing success rate
- More experience (flight number) → higher success

Business Implications

- Competitor can bid ~$62M if landing predicted successful
- Competitor should bid ~$150M if landing likely to fail
- Model gives 87.5% confidence to make this decision
- Focus on KSC LC-39A bids for maximum margin
- Payload range 2000–5000 kg is commercially optimal
- Model can be updated with new launches as data grows

Conclusion: Using publicly available data from the SpaceX API and Wikipedia, we successfully built a Decision Tree model that predicts Falcon 9 first-stage landing success with 87.5% accuracy. This enables competitive cost estimation for launch bids against SpaceX.

GitHub: https://github.com/Arijit-droid/IBM-DataScience-Capstone-SpaceX | IBM Data Science Capstone Project